

A Solution for A Disaster Recovery Service System in Multi-cloud Environment

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Abstract—The current enterprise information architecture is becoming more complex. As a result, user business data is lost or destroyed, services are interrupted, and systems crash, all of which harm users' businesses. These factors include equipment failure, network attacks, and human error. Data disaster recovery has become an important method to ensure the sustainability of enterprise business. It is an inevitable trend toward disaster recovery services and technologies that "business in the cloud, cloud disaster recovery, and multi-cloud disaster recovery." Traditional disaster recovery services systems and single-cloud disaster recovery services systems are unsatisfied with users' current demands for business system deployment in multi-cloud environments, cloud resource sharing, and cross-cloud disaster recovery. We discuss the structure, composition, and functional ideas of a solution for a disaster recovery service system in multi-cloud environments. To provide cross-cloud disaster recovery management service in a multi-cloud environment with full-stack virtualization in response to user business requirements, it combines conventional disaster recovery technology, blockchain distributed multi-copy data storage and security audit, and system security state awareness.

Keywords—Data disaster recovery, multi-cloud disaster recovery, virtualization, blockchain, state awareness

I. INTRODUCTION

Data disaster recovery is essential for the smooth running of corporate systems and is crucial for preserving data privacy, consistency, availability, and integrity. At the moment, cloud disaster recovery is gradually supplanting traditional data disaster recovery. Government agencies, businesses, and institutions are increasingly choosing to use virtualization technology to create cloud data backup, business sustainability assurance, and disaster recovery services through cloud disaster recovery services offered by cloud service providers.

However, the constant reported failures of cloud services remind us once again that while cloud services provide users with convenient information services, they also introduce new data security issues and business sustainability problems. When the business is deployed to the cloud centrally, a failure or disaster in the cloud data center will have more seriously effects as the user's business grows and a large amount of data aggregated. To cope with natural security risks and network attacks in data centers, cloud service providers build multiple geographically distributed data centers to achieve mutual data backup and disaster recovery management, service switching between multi-data centers. There are still two unresolved

issues. On the one hand, multiple data centers of the same cloud provider often use the same hardware and software technical architecture and operation and maintenance management system. The homogeneity may lead to synchronous failure of multi-data centers due to design flaws. On the other hand, users prefer to have their business run on different cloud service platforms and migrate freely between different clouds. However, due to technical barriers and commercial competition among different cloud platforms, the same user's resources cannot shared and used, the business cannot migrated freely, and user's data cannot used across clouds.

Therefore, from the perspective of shared use of cloud resources, unified management, and business disaster recovery, it is a better solution to use different cloud service platforms to build a cross-cloud multi-data center and multi-cloud environment disaster recovery service system, shield the differences between clouds, provide users with a unified resource management service system and intelligent disaster recovery backup service system, and avoid the unreliability of a single cloud service system and various restrictions on users, which is also an inevitable trend of future disaster recovery technology development.

In this paper, we propose a solution of multi-cloud disaster recovery business system (MCDR for short) for the disaster recovery needs of multi-cloud and heterogeneous cloud environments. MCDR realizes the unified management of user resources, business systems, user data and services in multi-cloud environments in the form of third-party centralized hosting, and provides users with unified disaster recovery and backup access services.

II. RELATED WORKS

Research on data disaster recovery backup for Multi-cloud involves studies on backup strategies [1], data de-duplication and bandwidth optimization [2], incremental synchronization [3], scheme selection, data consistency, and security assurance. Metastorage [4] selects the appropriate cloud node for data backup based on the response time of each cloud node. Cachin et al. [5] conducted a study from the perspective of consistency, integrity, and security assurance of backup data and designed a fault-tolerant protocol and a secure access control protocol in cloud mode. The DRCloud system proposed by Gu Y et al. [6] considers how to balance the reliability, low cost, and fast recovery of disaster recovery from the perspective of scheduling optimization strategy, etc. X. Li et al.[7] research the problem of redundancy and traffic distribution of disaster recovery data in cloud data centers and

construct a traffic ratio-constrained backup transmission model to achieve bandwidth-efficient and load-equitable disaster backup under redundancy and capacity constraints. Subsequently, X. Li et al.[8] further, construct capacity-constrained multicast backup models from the perspective of network resource management and multicast traffic aggregation capability, and combine ant colony optimization algorithms to adjust the ratio of bandwidth allocation for multiple backup transmissions with different demands to improve the flow rate. Abualkishik A Z et al.[9] studied from cloud disaster recovery strategy and compared it with traditional disaster recovery. Min Liang et al.[10] studied the application-level disaster backup fault detection problem and predicted the arrival time of cross-domain heartbeat information by particle swarm optimization-based back propagation neural network (PSO-BP), using actively assisted detection (AD) to improve the correctness of fault detection. Shanwen Yi et al.[11] carried out research on the problem of transmission energy consumption during disaster backup of long distance and massive data, using multi-objective reinforcement learning to minimize the number of forwarding devices in the middle of the transmission process, reduce the transmission time and decrease the energy consumption. The above-mentioned studies focus on the technical aspects of disaster recovery and only cover the technical implementation of disaster recovery services, but do not study the cross-cloud disaster recovery service system and technical architecture from the macro level.

III. MCDR ARCHITECTURE AND APPLICATION DEPLOYMENT

Facing the huge differences in virtualization technology architecture and service management of cloud platforms in multi-cloud heterogeneous environments, we need to treat multi-cloud disaster recovery as a technical service system from the perspective of third-party services to provide unified services to users and shield the underlying differences. The application deployment structure of MCDR shows in Figure 1. In the figure, we assume that users use the services of Ali Cloud, Tencent Cloud and Huawei Cloud to build a multi-cloud disaster recovery service system. The multi-cloud disaster recovery service system receives disaster recovery service requests from users in a centralized management manner, converts the requests into service requests from different clouds, helps users complete task requests using specific cloud services, receives service result information from cloud service feedback, and manages distributed cloud storage of cloud resources and disaster recovery data.

MCDR uses resource access API interfaces provided by different cloud service providers and uses multi-threaded programming to achieve concurrent execution of multiple services. The system includes several component modules. The system composition structure shows in Figure 2 below.

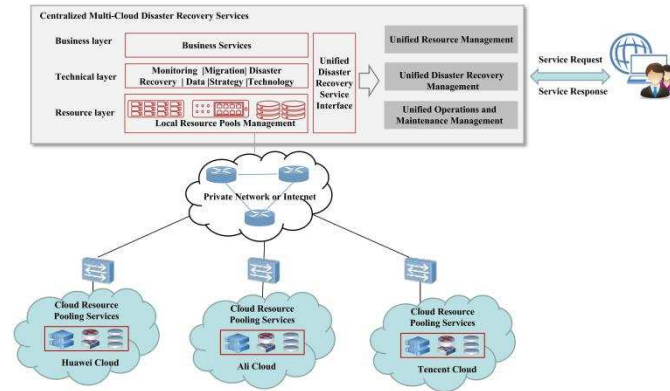


Figure 1. Application deployment structure of MCDR

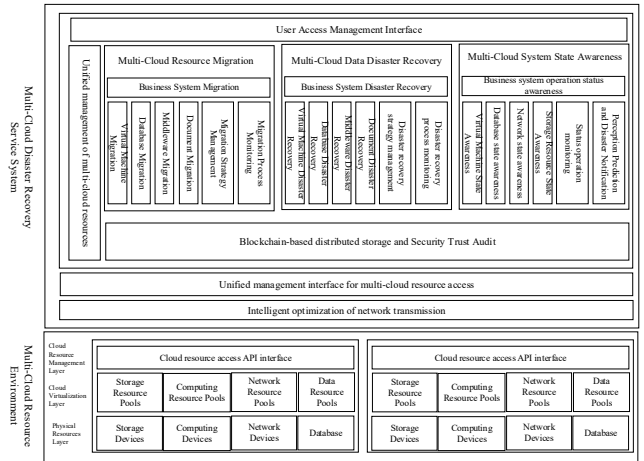


Figure 2. Architecture of MCDR

IV. ARCHITECTURE DESCRIPTION OF MCDR

A. Intelligent Optimization of Network Transmission

One of the key issues in implementing cross-cloud disaster recovery for multi-cloud environments is to solve the problem of real-time transmission of massive data over long distances. Data transmission performance is optimized using multi-agent, parallel stream transmission. First, the data is classified according to the disaster recovery performance requirements of user business data. High weights are assigned to critical business data or data with high real-time requirements for priority transmission. Secondly, when performing the transfer task, several transfer agents are assigned in multi-clouds to divide the data transfer workload into several data fragments, and the transfer tasks are assigned reasonably according to the bandwidth metrics of the transfer agents. If the available bandwidth falls below the threshold, the cloud-based transfer agent is added dynamically to cope with the sudden transfer workload. A feedback control strategy is used in the data transmission process to intelligently adjust and optimize the bandwidth resources occupied by the tasks according to the time delay of data transmission between transmission agents.

B. Unified management interface for multi-cloud resource access

In a multi-cloud environment, users' resources (including virtual machines, databases, storage, and other resources) are distributed in different clouds. To solve the problem of not being able to provide cross-platform disaster recovery services to users due to technical differences and resource isolation among cloud service providers, it is necessary to

build a unified resource access interface on top of different platforms for adapting the technical implementation of different cloud platforms, shield the differences in service providers, technical systems, and resource architectures of the underlying cloud resources, provide a unified API interface for users to use the upper multi-cloud disaster recovery service system, and realize service access, format conversion, and message communication for differentiated cloud platforms.

C. Distributed Storage and Security Trust Audit

In traditional disaster recovery solutions and single cloud disaster recovery solutions, backup data is stored in a centralized manner. Once the storage server is compromised by an attacker, it poses a serious security risk to the user's backup data. With data backup using cloud services, the owner of the data and the storage manager are separated, and the storage manager (cloud service provider) can leak the user's backup data to a third party for financial gain without the owner of the data (user) knowing anything about it.

In the traditional centralized disaster recovery data storage, storage servers are vulnerable to cyber-attacks. In the single-cloud data disaster recovery approach, the ownership of data and storage management are separated. The storage manager (cloud service provider) can leak the user's backup data to a third party for financial gain without the owner of the data (user) knowing it at all. The about mentioned issues can be resolved with blockchain technology. As an autonomous and trustworthy application model, blockchain uses distributed storage, smart contracts, cryptography, and consensus mechanisms to provide tamper-proof and traceability. Its features are excellent for meeting the needs of data backup and disaster recovery in a multi-cloud environment.

We consider the user business side (the actual owner of the original data), MCDR (acting as a trusted audit server), and the cloud platform storage nodes as nodes in the blockchain network. User requests for data backup and recovery are subject to security and compliance audits by MCDR. User requests that pass the audit will be finally executed. We treat user disaster recovery requests, security audits of MCDR, and storage processes of cloud platform storage nodes as data transaction activities in the blockchain. For data storage, the original disaster data is sliced. Each slice is encrypted using a symmetric encryption algorithm. The ciphertext slice is calculated the digest value using a Hash algorithm. The digest value is digitally signed using the user's private key. Then the slice ciphertext, digest, and digital signature are treated as one transaction data and stored in the storage nodes of the cloud platform in a multi-copy manner using a threshold secret sharing scheme. Each node stores a certain number of ciphertext slices, and the same slice is stored on multiple nodes. This approach makes it possible to guarantee user data reliability and integrity even if some storage nodes fail. After receiving the user ciphertext data slice, the storage node verifies the user signature using the user public key, checks the integrity of the data using the hash value, and then stores the data. To save storage resources on the blockchain and reduce the computational cost, the storage records on the blockchain hold only the metadata of the disaster recovery data. Metadata is the management index of disaster recovery data, through which you can retrieve the actual storage location, data volume, data information summary, etc. During data recovery, the user business side reads metadata through blockchain and completes retrieval of ciphertext data, parallel

download, signature verification, integrity check, ciphertext decryption, and merging data slice to obtain the original data.

This solution uses blockchain technology for secure storage of multiple copies of data, secure access auditing, data tampering, and loss prevention functions in a decentralized manner. It can also guarantee the reliability and integrity of backup data in the event of physical failure of some storage nodes in the cloud.

D. Unified management of Multi-Cloud Resource

This layer provides unified management of resources scattered in homogeneous or heterogeneous cloud platforms or regions. It uses the adaptation capability to heterogeneous cloud environments provided by the multi-cloud resource access unified management interface to integrate scattered user resources into a unified resource pool for on-demand allocation, achieve balanced and optimized management of resource usage efficiency, and provide consistent user management, authority allocation, security auditing, and unified cloud resource application, definition, query, and operation services.

E. Multi-Cloud Resource Migration

Multi-cloud resource migration requires all kinds of virtual resources related to user business systems, including virtual machines, databases, middleware, files, etc., to be migrated from local to cloud or freely migrated between different availability zones in the same cloud or between availability zones in different clouds according to the set migration strategies.

The content of data migration is mainly the business application data and configuration files directly related to the user's business. the migration source, destination, time policy, and security policy are necessary before creating the migration task, monitoring the migration process in real-time, and producing the performance report of the task. This allows the migration task to be queried and tracked at any time. Combined with network transmission optimization requirements, migration data needs to be combined with data segmentation, compression, de-duplication, segmented transmission, breakpoint sequential transmission, data merging, integrity testing, etc., and reduce migration data volume, achieve migration efficiency and data reliability, and reduce the impact of migration services on user business system.

F. Multi-Cloud Data Disaster Recovery

The traditional local data disaster recovery and same cloud disaster recovery services are not conducive to resource sharing and data reliability assurance among different clouds. Multi-cloud data disaster recovery manages backup and recovery of user's business system resources (including virtual machines, databases, middleware, and files) in the same or different clouds. In case of business data loss or abnormality at the source end, users can restore the data of the business system to the specified cloud storage based on the backup data. To share storage resources in different clouds and ensure data reliability, the user data of multi-cloud backup is stored in different clouds using distributed and multi-copy approach for decentralized storage, and metadata is used to manage and index the storage backup. When recovering data, you can retrieve the storage location through metadata, perform consistency detection and multi-source data recovery for

different copies of the same data, and improve the efficiency of data recovery and data reliability.

G. Multi-Cloud System state Awareness

Traditional disaster recovery technology is a reactive remediation model, which is inevitably affected by the timeliness and integrity of data backup and the backup method and volume of backup.

Multi-cloud system state awareness adopts the idea of disaster preparedness defense and provides an environment-based, dynamic, and global perspective of security risk prediction and defense capability for the system by monitoring and tracking the factors affecting the user's business system in real-time, and predicting the possibility of disaster events. It includes three primary functions, state data collection, security state modeling, and intelligent disaster prediction. Status data collection collects operational metrics for user-selected business systems and IT resource types, including virtual machines, networks, storage, and applications. Security state modeling from four levels: operating system, user business system, disaster recovery business, and monitoring components to set the security state metrics of users' business system, classify and quantify the state data, set evaluation thresholds, alarm policies, and weight coefficients of metrics, normalize and weight the data, and perform threshold comparison and alarm processing, etc. Intelligent disaster prediction quantifies security risks and baseline analysis of current system state data, helps users understand whether applications or services are in normal and stable status, automatically assesses state security risks, predicts the probability of assets suffering from security risks, and uses this as a basis for early disaster awareness and warning, and prepares for automated disaster recovery and backup and business migration.

H. User Access Management Interface

User access management interface provides users with a unified management UI interface for multi-cloud environment and multi kinds of business resources. The module adopts front and back-end separation technology and uses Vue framework technology to develop a graphical user access interface, providing users with a unified graphical user form for implementing resource management, resource migration, business disaster recovery, resource monitoring and disaster recovery awareness in multi-cloud environments. Users no longer need to focus on the underlying implementation of cloud environments and differential services, but focus on business and service management, business logic orchestration and monitoring.

V. CONCLUSION

In view of the development of disaster recovery technology direction of "upstream cloud disaster recovery, multi-cloud disaster recovery, intelligent disaster recovery", this paper describes the structure, composition, and functional ideas of a multi-cloud intelligent disaster recovery service system solution, which combines traditional disaster recovery technologies to realize cross-cloud disaster recovery management in a multi-cloud environment with full-stack

virtualization for users' business needs in a heterogeneous cloud environment.

With the rapid growth in data volume and the continued increase in data value, the demand for data disaster recovery using cloud-based virtualization platforms continues to grow. To build a standardized and elastic virtualized resource platform, realize the unified management and allocation, and scheduling of resources, meet the hierarchical disaster recovery needs of users' business applications, and provide users with unified disaster recovery services, it is not only necessary to consider the compatibility issues of vendors, versions, hardware and software, implementation technologies, cost factors, time factors, legal factors, but also the security issues brought by the separation of resource ownership and management, which has become a long-term research issue in the development of cloud disaster recovery technology and deserves our continuous attention.

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REFERENCES

- [1] Zhang J, Zhang N. Cloud Computing-based Data Storage and Disaster Recovery [C]. Proceedings of the 2011 International Conference on Future Computer Science and Education, IEEE Computer Society, 2011:629-632.
- [2] Tan Y, Jiang H, Feng D, et al. CABdedupe: A Causality-Based Deduplication Performance Booster for Cloud Backup Services [C]. Proceedings of the 2011 IEEE International Parallel & Distributed Processing Symposium, IEEE Computer Society, 2011:1266-1277.
- [3] Li W, Yang Y, Yuan D. A novel cost-effective dynamic data replication strategy for reliability in cloud data centres [C]. Proceedings of the IEEE 9th International Conference on Dependable, Autonomic and Secure Computing, IEEE, 2011:496-502.
- [4] Bermbach D, Klems M, Tai S, et al. MetaStorage: A Federated Cloud Storage System to Manage Consistency-Latency Tradeoffs [C]. Proceedings of the 2011 IEEE 4th International Conference on Cloud Computing, IEEE Computer Society, 2011:452-459.
- [5] Cachin C, Haas R, Vukolic M. Dependable storage in the Intercloud [J]. IBM Research, 2010, 3783:1-6.
- [6] Gu Y, Wang D, Liu C. DR-Cloud: Multi-cloud based disaster recovery service[J]. Tsinghua Science and Technology, 2014, 19(1): 13-23.
- [7] X. Li, H. Wang, S. Yi, X. Yao, F. Zhu and L. Zhai, "Redundancy-Guaranteed and Receiving-Constrained Disaster Backup in Cloud Data Center Network," in IEEE Access, vol. 6, pp. 47666-47681, 2018, doi: 10.1109/ACCESS.2018.2859427.
- [8] Li X, Wang H, Yi S, et al. Cost - efficient disaster backup for multiple data centers using capacity - constrained multicast[J]. Concurrency and Computation: Practice and Experience, 2019, 31(17): e5266.
- [9] Abualkashik A Z, Alwan A A, Gulzar Y. Disaster recovery in cloud computing systems: An overview[J]. International Journal of Advanced Computer Science and Applications, 2020, 11(9).
- [10] Liang M, Gao F, Shi M. A failure detection method of remote disaster recovery and backup system[C]//ITM Web of Conferences. EDP Sciences, 2022, 47: 01002.
- [11] Yi S, Li X, Wang H, et al. Energy - aware disaster backup among cloud datacenters using multiobjective reinforcement learning in software defined network[J]. Concurrency and Computation: Practice and Experience, 2022, 34(3): e6588.